

PSG Institute of Technology and Applied Research

Computer Science and Business Systems
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Autism Spectrum Disorder (ASD) Early Detection with Explainable AI for Infants, Children & Teens

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Introduction

- Autism spectrum disorder (ASD) is a neurodevelopmental condition affecting communication, social interaction, and behavior.
- Early detection improves long-term cognitive, behavioral, and social outcomes.
- Traditional diagnosis relies on behavioral assessment by clinicians → subjective and time-consuming.
- Ai-based systems help automate early risk detection using gaze patterns, facial cues & behavioral markers.
- This project focuses on **three age groups**:
  - Infants (0 – 3 years)
  - Children (4–12 years)
  - Teens (13 – 17 years)

Problem Statement

- Early ASD indicators vary significantly across age groups (infants, children, teens).
- Most solutions are designed for a specific age group and depend on clinical data
- There is **no unified ai-based system** that:
  - Analyses behavioral & gaze patterns across all three developmental stages,
  - Provides explainable reasoning behind predictions.

Objectives

- To develop an ai-based system for early ASD risk detection in
  - Infants using gaze behavior analysis.
  - Children using behavioral questionnaire analysis.
  - Teens using cognitive response and game-based interaction analysis.
- To Apply explainable AI (SHAP) for interpreting predictions.
- To Build age-wise classification models with segmented datasets.

Literature Review

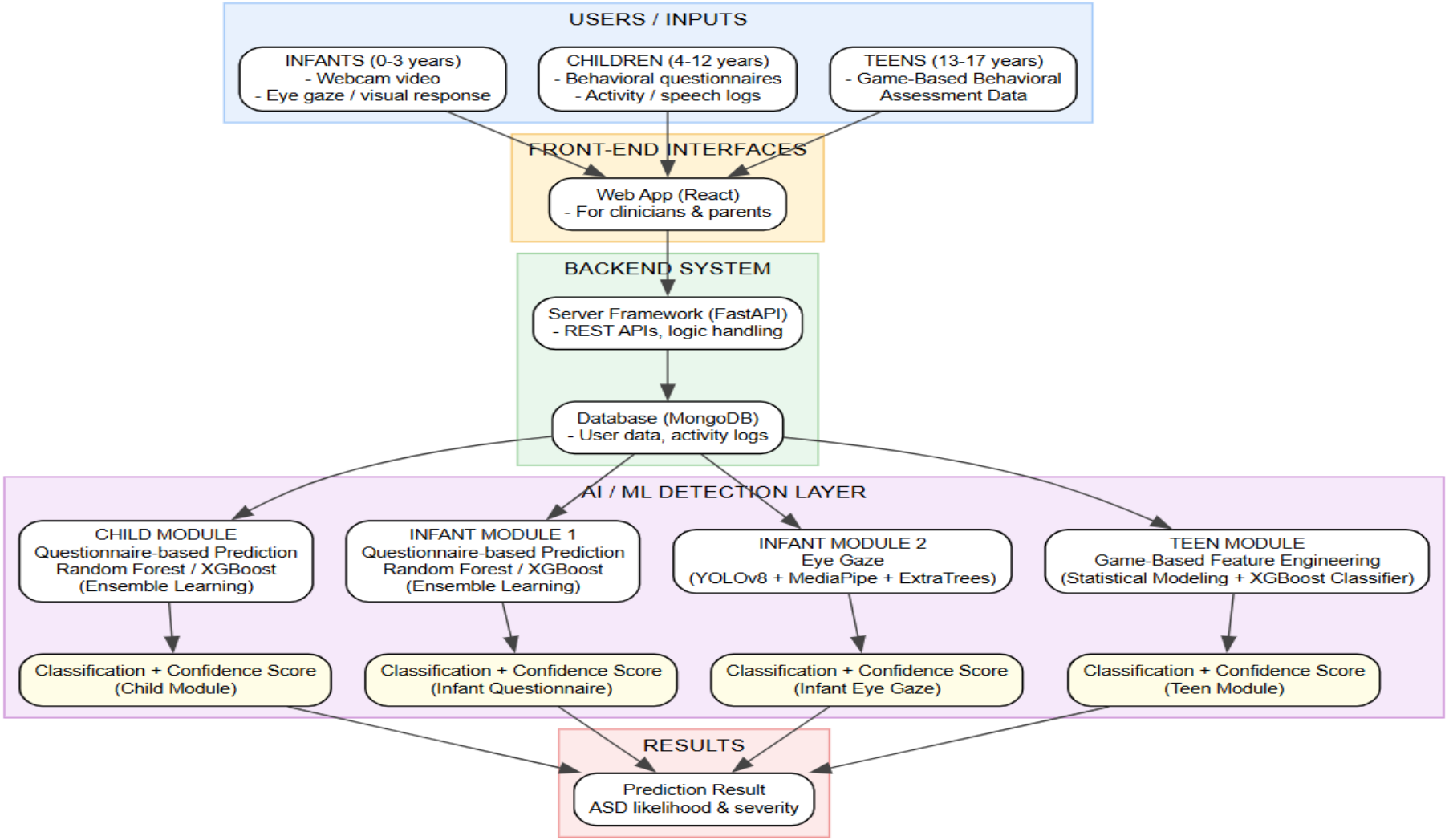
Sno	Title, Author & Year	Inferences
1.	"Detection of Autism Spectrum Disorder in Children Using Federated Machine Learning Techniques",M. S. Farooq, M. Riaz, and A. Ali (2023)	• Enables ASD detection using decentralized data without sharing sensitive information • Improves data privacy and security in healthcare applications
2.	"An Evaluation of Machine Learning Approaches for Early Autism Detection",R. A. Rasul et al. (2024)	• ML models effectively classify ASD using behavioral and clinical data • Provides high accuracy and supports early diagnosis
3.	"Using Machine Learning for Motion Analysis to Early Detect Autism Spectrum Disorder",R. Simeoli and L. Rossi (2024)	• Motion patterns can reveal early behavioral differences in ASD individuals •ML models can analyze movement data for early detection

Research gap identified:

- No single work integrates **all 3 age groups**, **real-time gaze extraction**, and **explainable AI (SHAP)** in one unified early detection system.

Proposed System

System Architecture



Proposed System

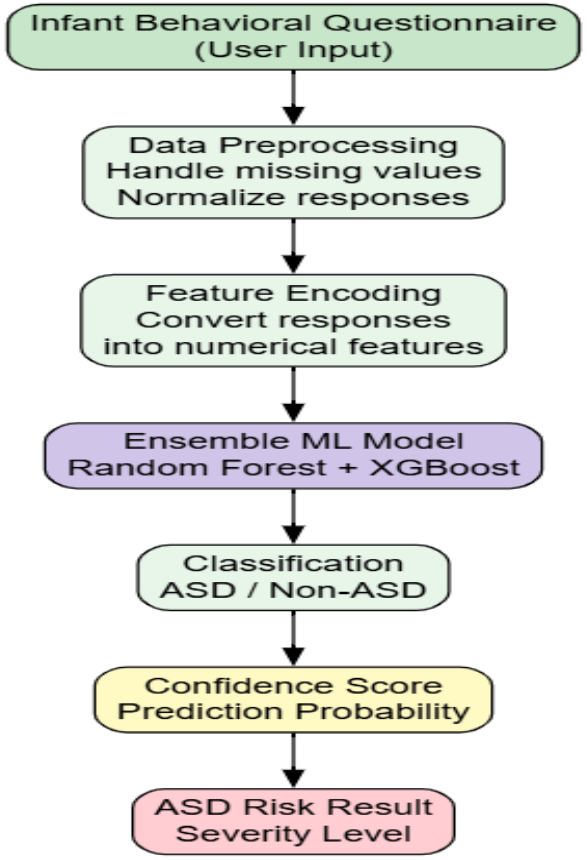
Workflow

- Data Input (Video / Questionnaire / Game Data)
- Preprocessing & Feature Extraction
- YOLO + MediaPipe Gaze Detection
- Behavioral & Cognitive Feature Engineering
- Age-wise ASD Classification
- SHAP Explainability
- Final Prediction & Report Generation

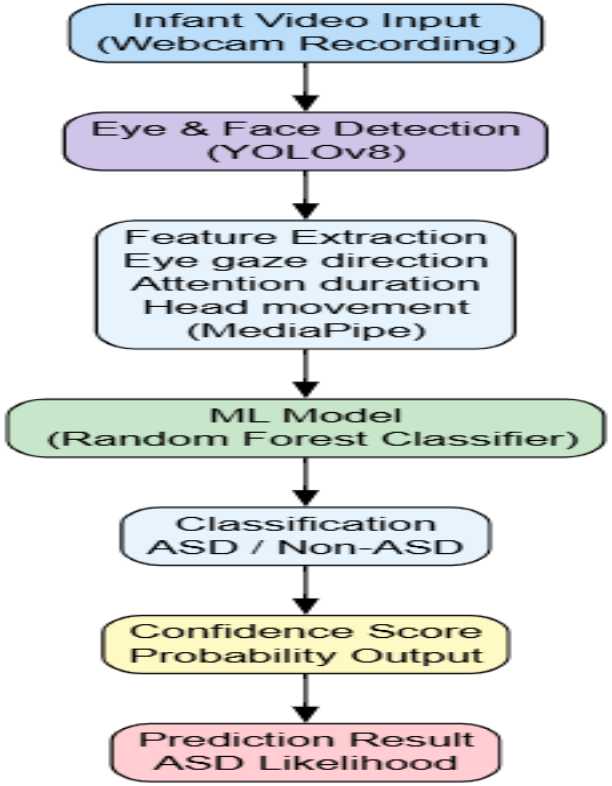
Proposed System

FlowChart

Infant behavioural module



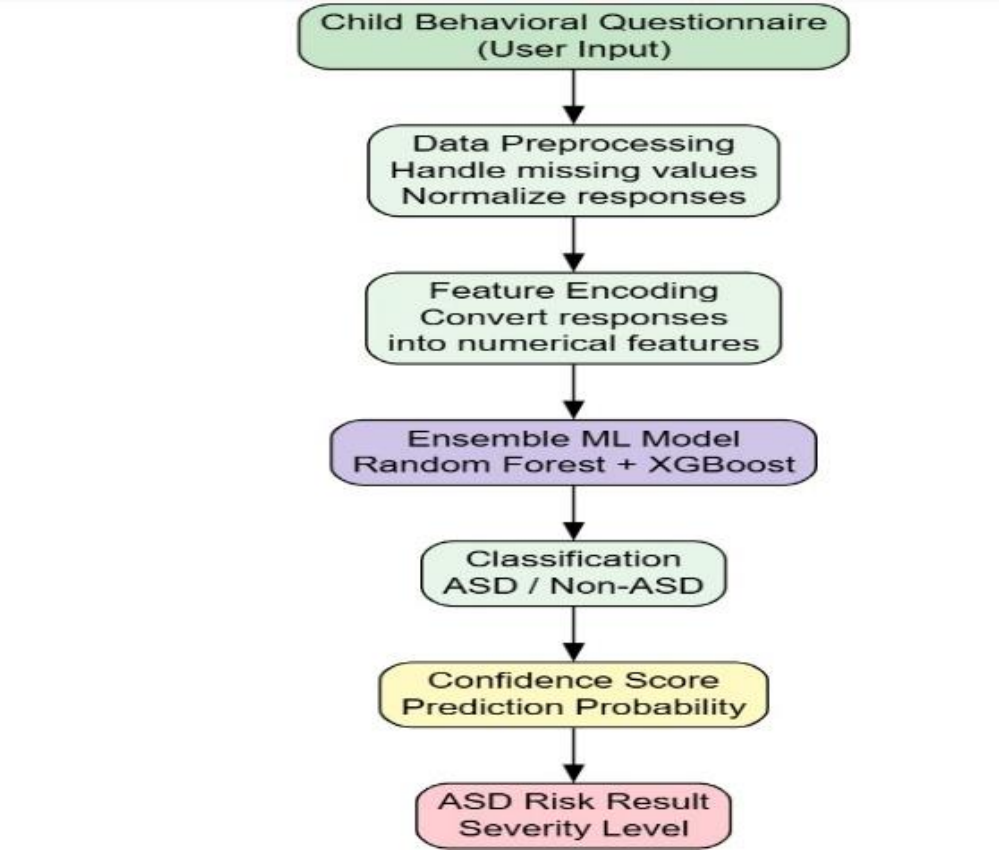
Infant gaze analysis module



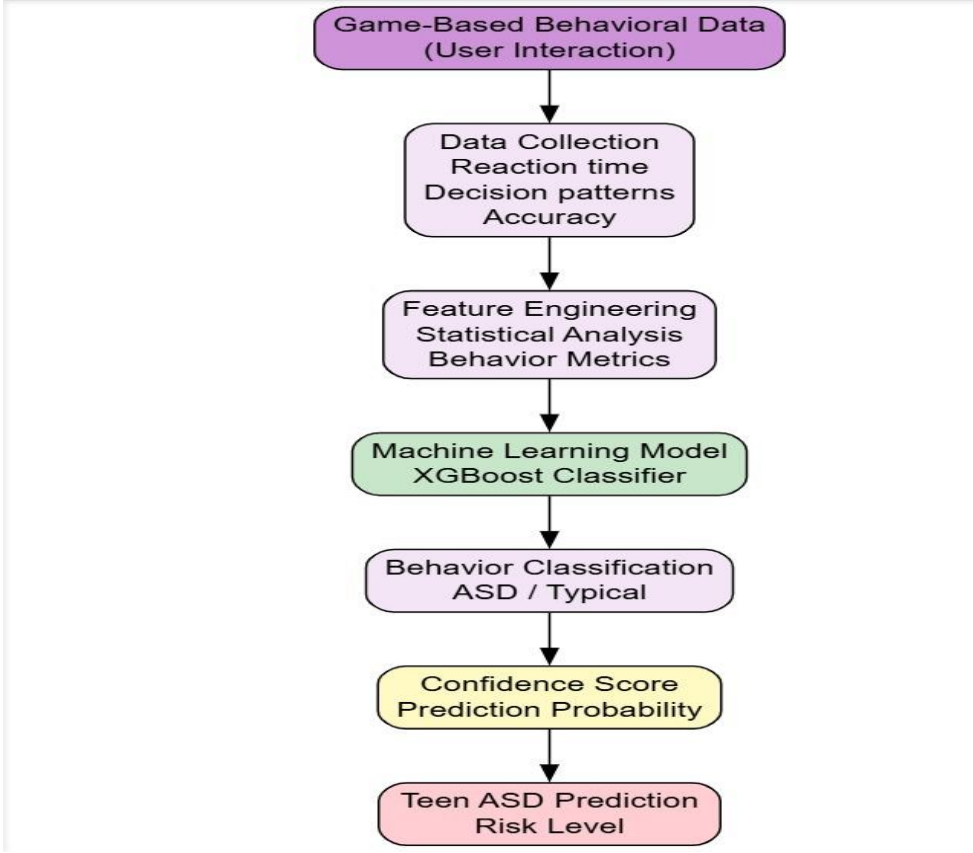
Proposed System

Flow Chart

Child behavioural module



Teen Cognitive Simulation module



Proposed System

Key models used

- **Yolov8** for eye & face detection
- **Mediapipe** for landmarks
- **Isolation Forest & KNN Imputation**
- **Xgboost & randomforest** for final classification
- **SHAP** for explainability

Dataset Used

- **Behavioral questionnaire dataset (children & infants):**
    - Behavioral screening data collected from kaggle, comprising approximately **2.2 lakh labeled questionnaire entries**
- **Infant eye-tracking dataset:**
    - Eye-gaze tracking dataset sourced from kaggle, containing approximately **1.5 lakh labeled infant gaze samples**
- **Teen game-based behavioral dataset**:
- Game-based behavioral assessment dataset from UCI repositories, consisting of **over 3 lakh interaction records** for teen cognitive analysis

Implementation

Modules implemented

1. Infant module

- Extracts gaze stability, facial reactivity, fixation time
- Detects unusual gaze avoidance patterns
- Tracks behaviour using AQ responses

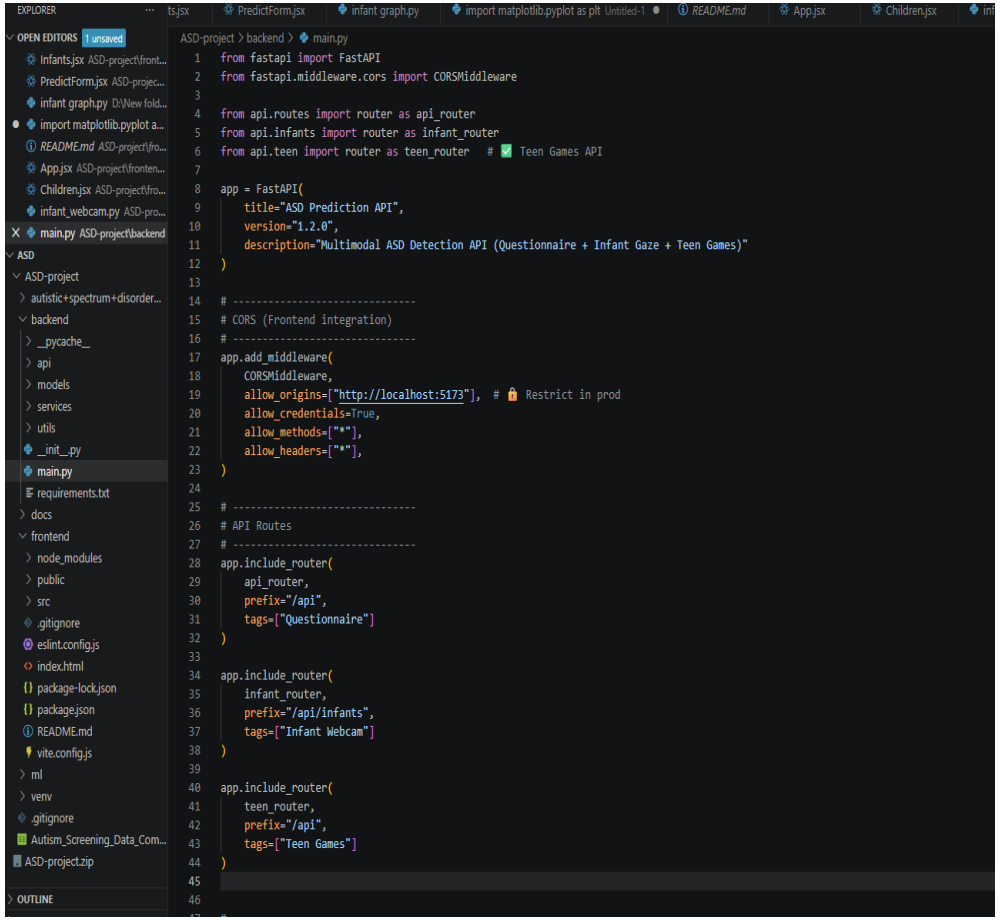
2. Children module

- Tracks social behaviour using questionaries
- Evaluate behavioural Patterns

3. Teen module

- Social-emotional cues and behaviour using game based activities
- Tracks reaction time, attention span, impulse behaviour

Implementation



```python
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware

from api.routes import router as api_router
from api.infants import router as infant_router
from api.teen import router as teen_router   # ✅ Teen Games API

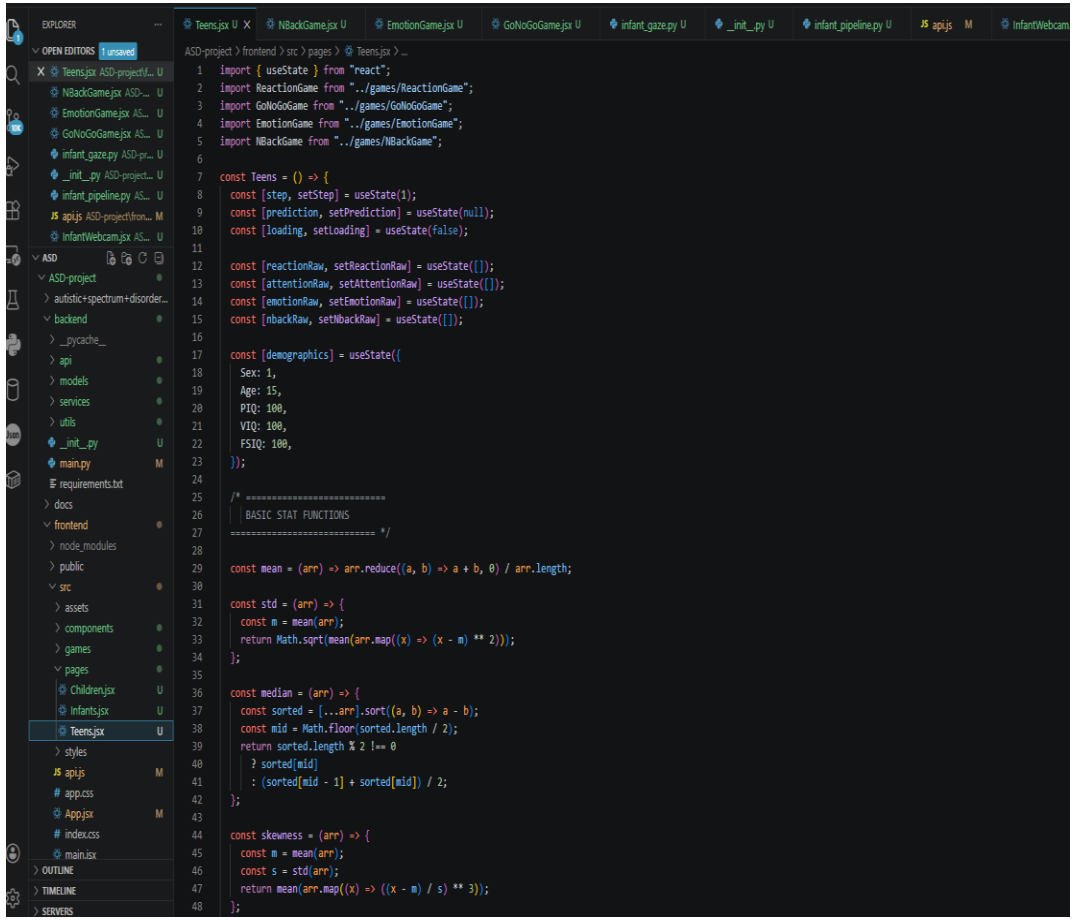
app = FastAPI(
    title="ASD Prediction API",
    version="1.2.0",
    description="Multimodal ASD Detection API (Questionnaire + Infant Gaze + Teen Games)"
)

# ------------------------------
# CORS (Frontend integration)
# ------------------------------
app.add_middleware(
    CORSMiddleware,
    allow_origins=["http://localhost:5173"],  # 🔒 Restrict in prod
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

# ------------------------------
# API Routes
# ------------------------------
app.include_router(
    api_router,
    prefix="/api",
    tags=["Questionnaire"]
)

app.include_router(
    infant_router,
    prefix="/api/infants",
    tags=["Infant Webcam"]
)

app.include_router(
    teen_router,
    prefix="/api",
    tags=["Teen Games"]
)
```



```javascript
import { useState } from "react";
import ReactionGame from "../games/ReactionGame";
import GoNoGoGame from "../games/GoNoGoGame";
import EmotionGame from "../games/EmotionGame";
import NBackGame from "../games/NBackGame";

const Teens = () => {
  const [step, setStep] = useState(1);
  const [prediction, setPrediction] = useState(null);
  const [loading, setLoading] = useState(false);

  const [reactionRaw, setReactionRaw] = useState([]);
  const [attentionRaw, setAttentionRaw] = useState([]);
  const [emotionRaw, setEmotionRaw] = useState([]);
  const [nbackRaw, setNbackRaw] = useState([]);

  const [demographics] = useState({
    Sex: 1,
    Age: 15,
    PIQ: 100,
    VIQ: 100,
    FSIQ: 100,
  });

  /* ==========================
     BASIC STAT FUNCTIONS
  ========================== */

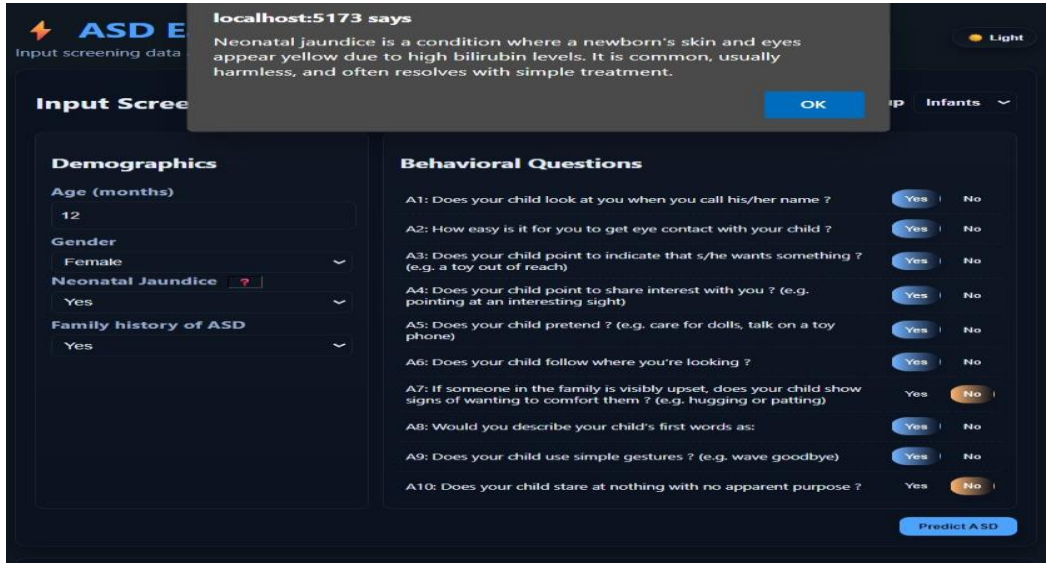
  const mean = (arr) => arr.reduce((a, b) => a + b, 0) / arr.length;

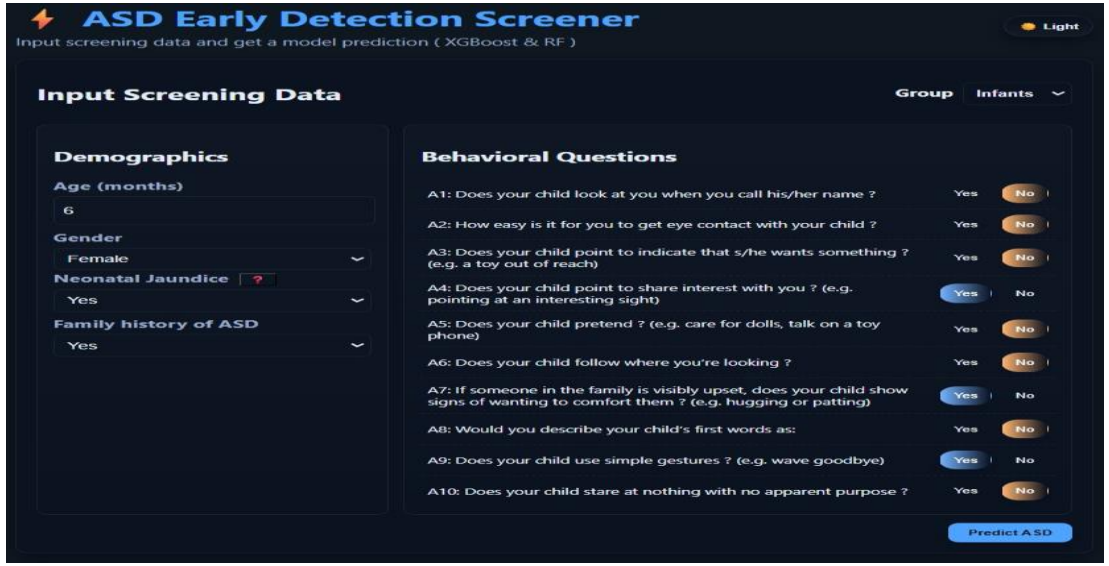
  const std = (arr) => {
    const m = mean(arr);
    return Math.sqrt(mean(arr.map((x) => (x - m) ** 2)));
  };

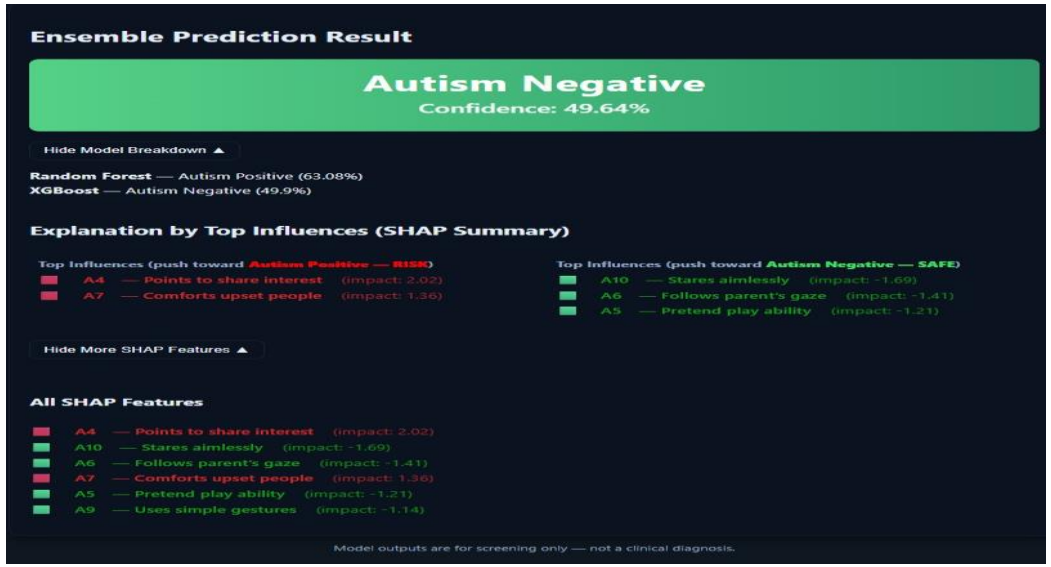
  const median = (arr) => {
    const sorted = [...arr].sort((a, b) => a - b);
    const mid = Math.floor(sorted.length / 2);
    return sorted.length % 2 !== 0
      ? sorted[mid]
      : (sorted[mid - 1] + sorted[mid]) / 2;
  };

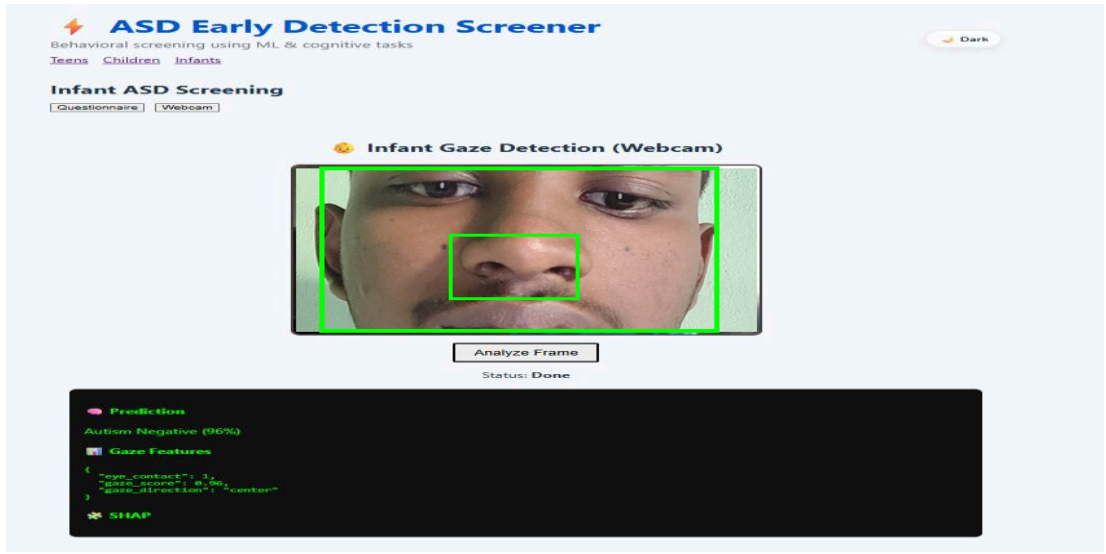
  const skewness = (arr) => {
    const m = mean(arr);
    const s = std(arr);
    return mean(arr.map((x) => ((x - m) / s) ** 3));
  };
```

Results & Discussions







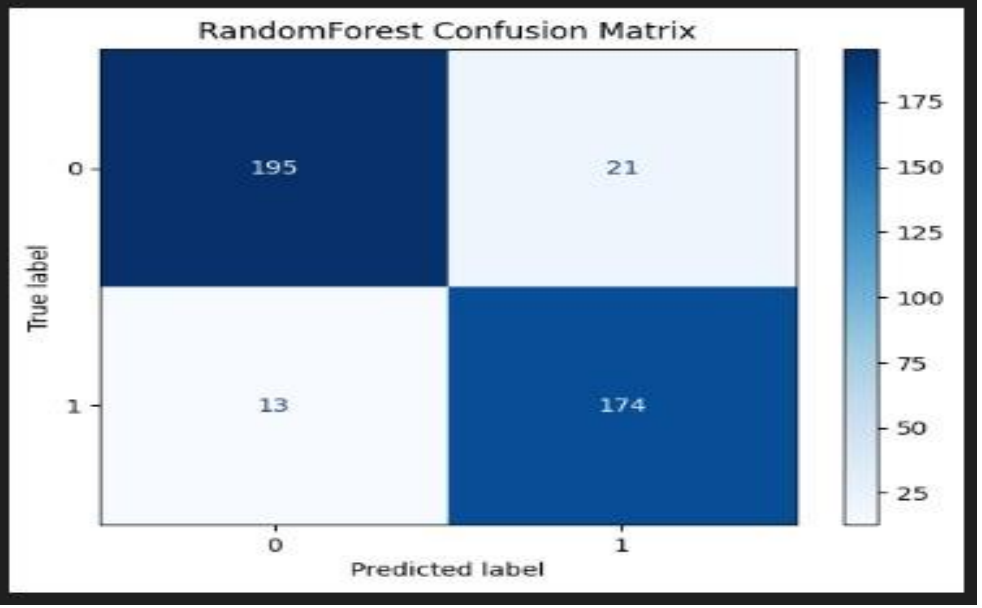


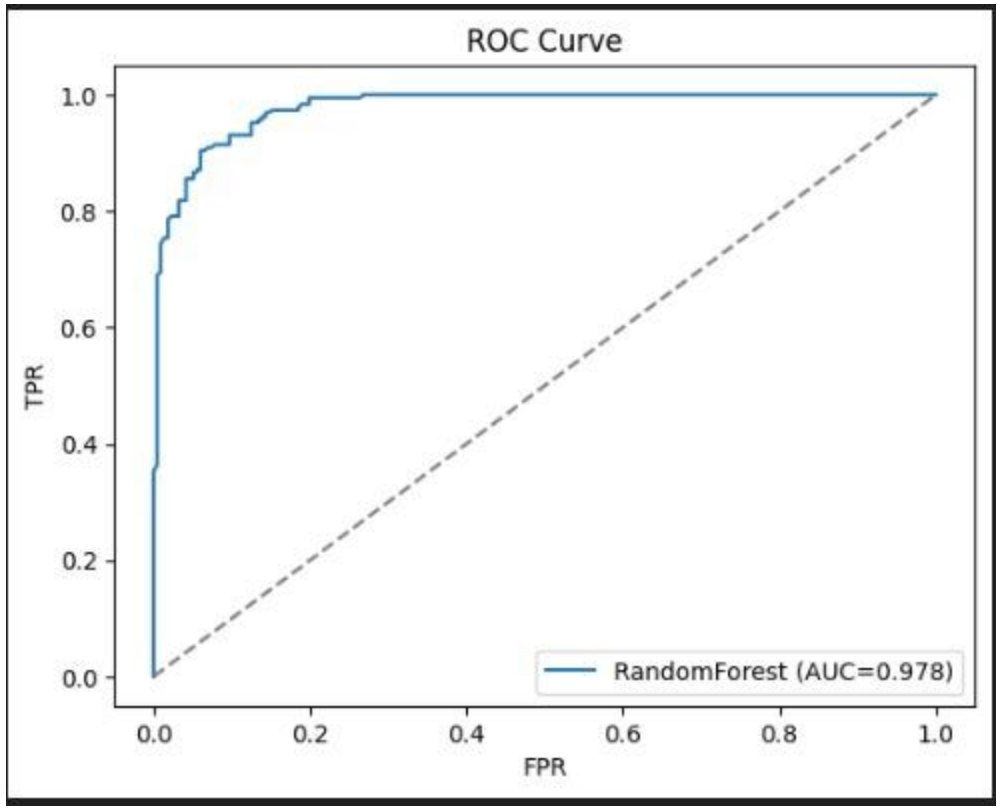
Results & Discussions

```
--- RandomForest ---
Accuracy: 0.9156327543424317
ROC AUC: 0.9783868092691622
              precision    recall  f1-score   support

           0       0.94      0.90      0.92       216
           1       0.89      0.93      0.91       187

    accuracy                           0.92       403
   macro avg       0.91      0.92      0.92       403
weighted avg       0.92      0.92      0.92       403
```

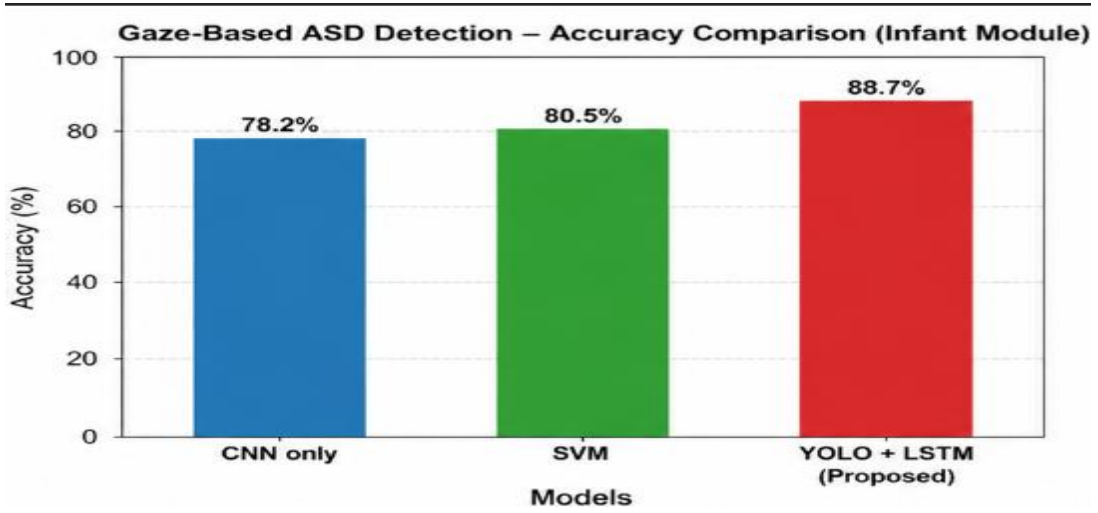


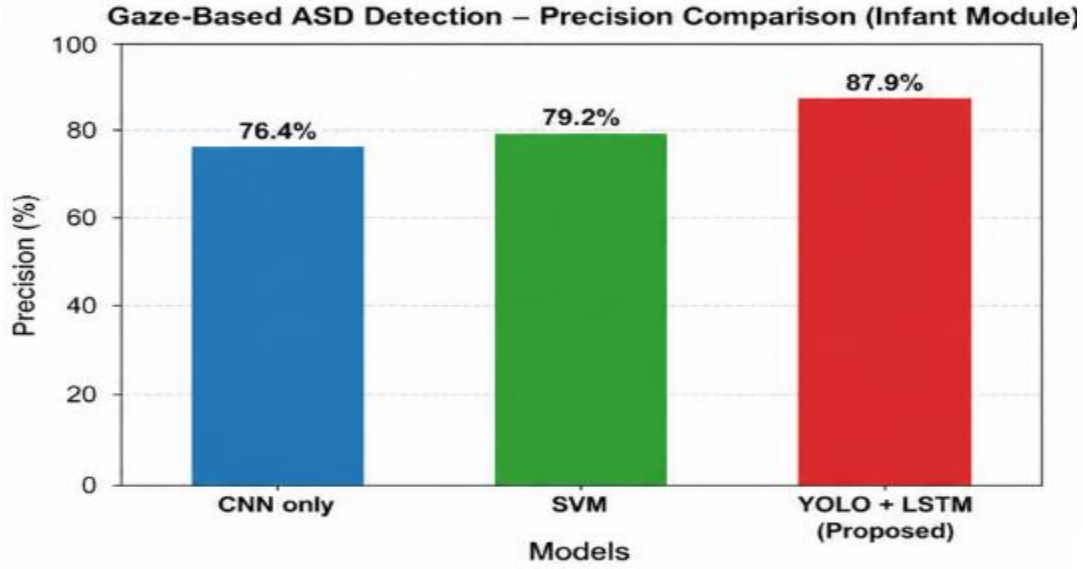


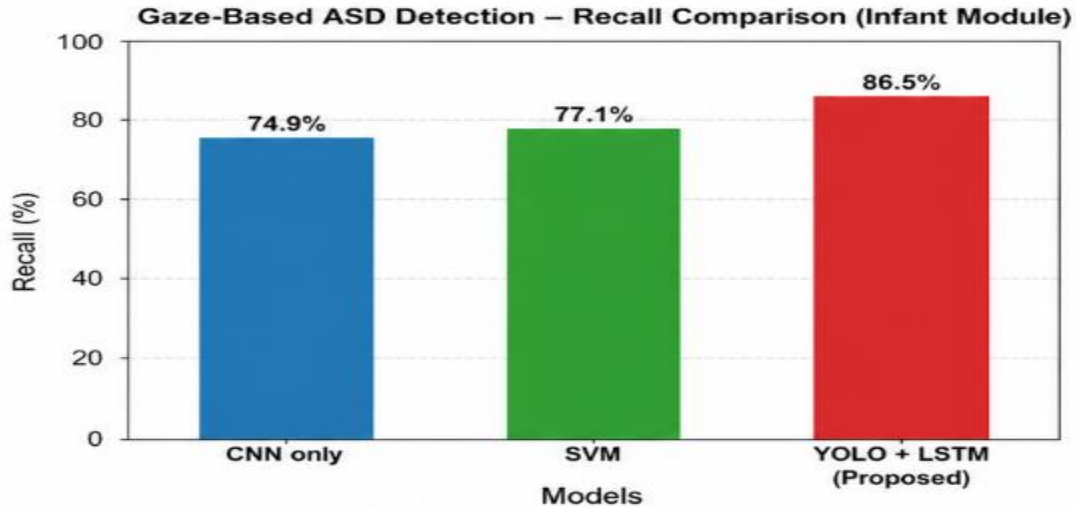
Results & Discussions

TABLE I *Gaze-Based ASD Detection Performance*

Model	Accuracy	Precision	Recall	F1-Score
CNN only	78.2%	76.4%	74.9%	75.6%
SVM	80.5%	79.2%	77.1%	78.1%
YOLO + LSTM (Proposed)	88.7%	87.9%	86.5%	87.2%



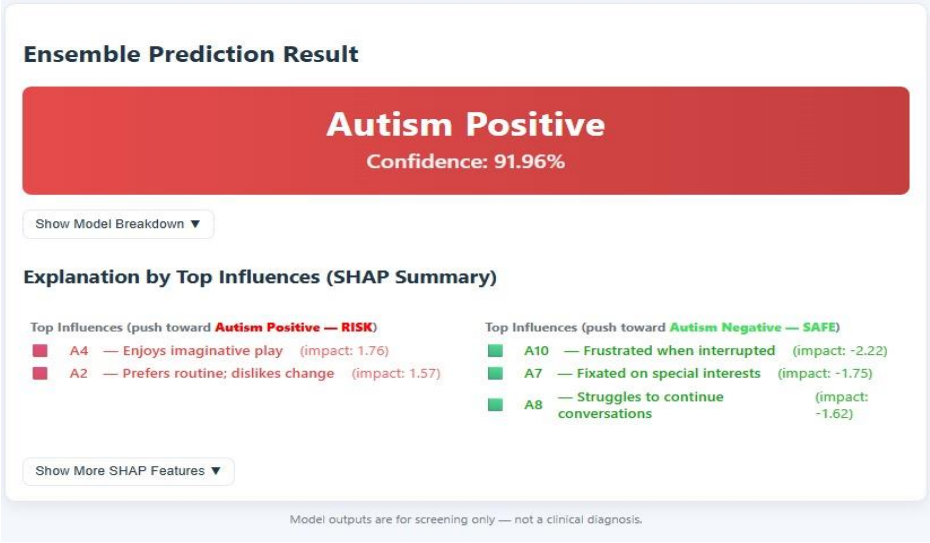


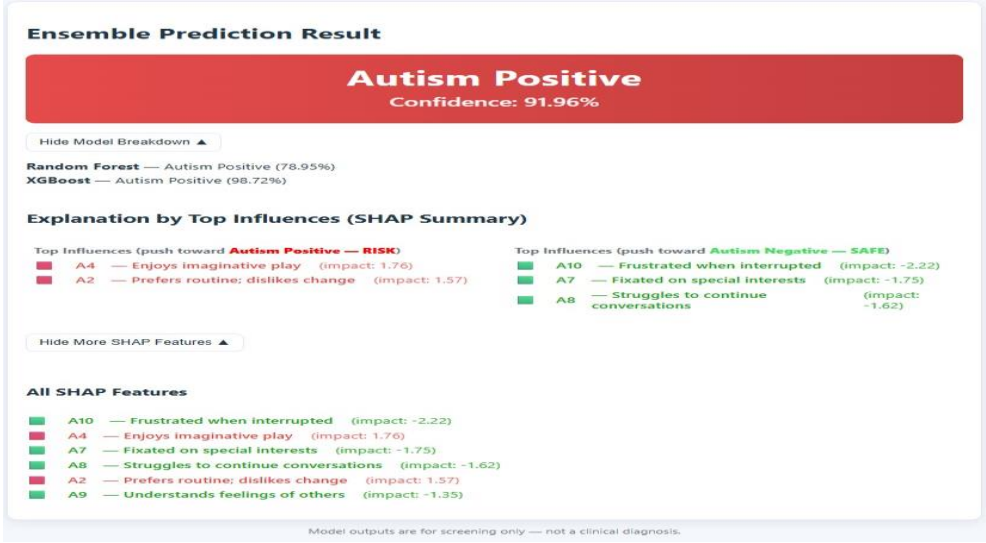


Results & Discussions

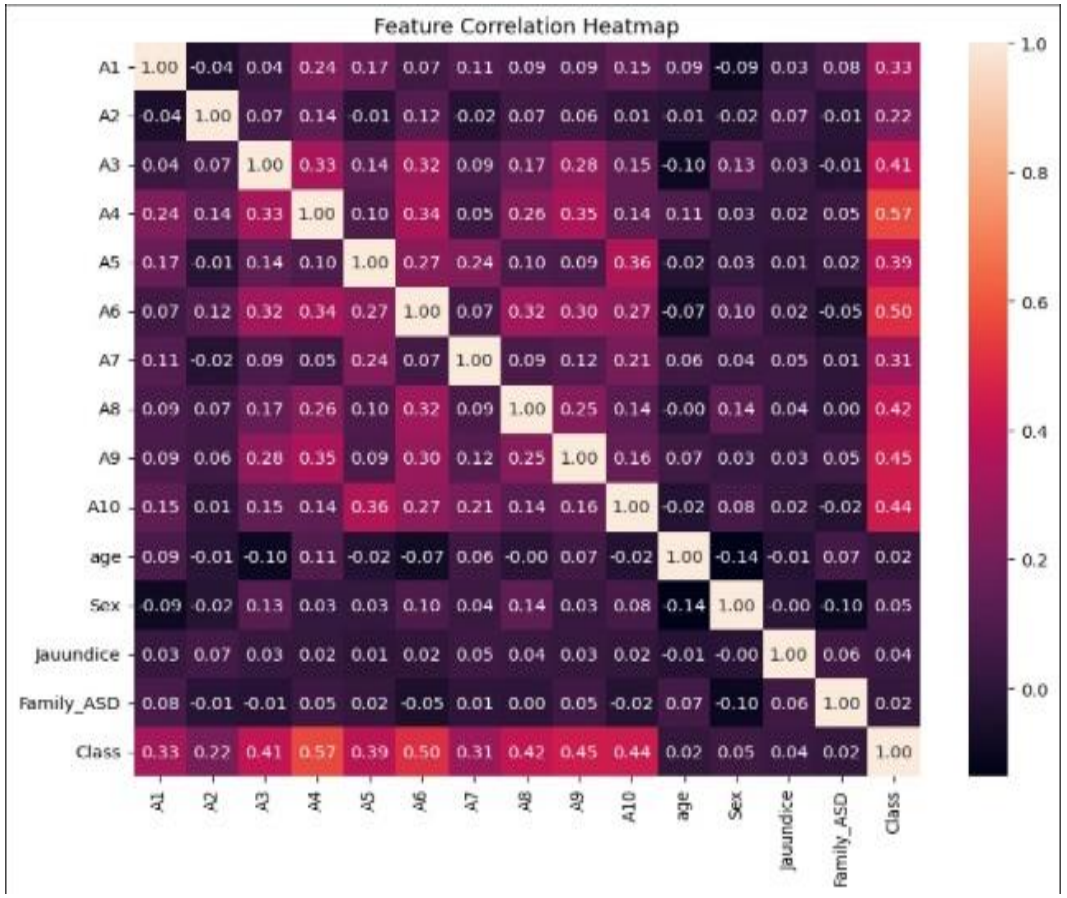


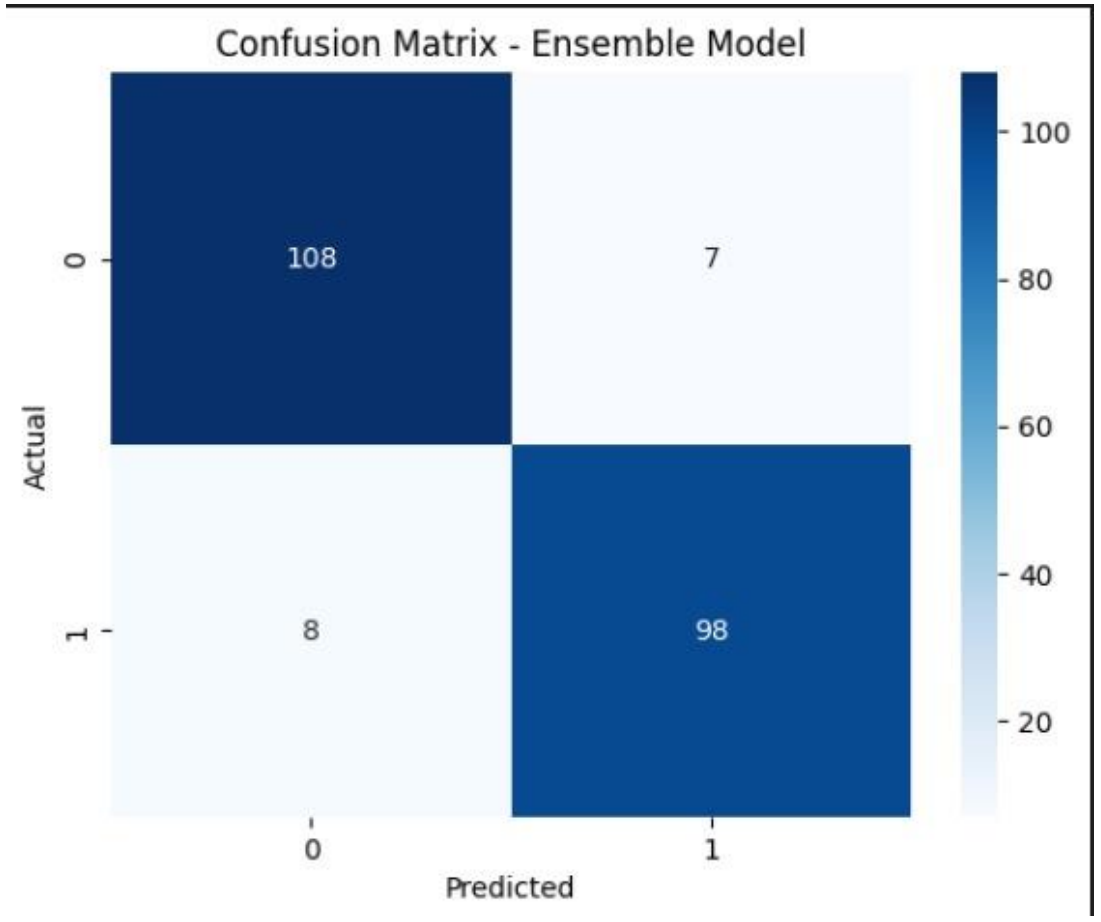






Results & Discussions

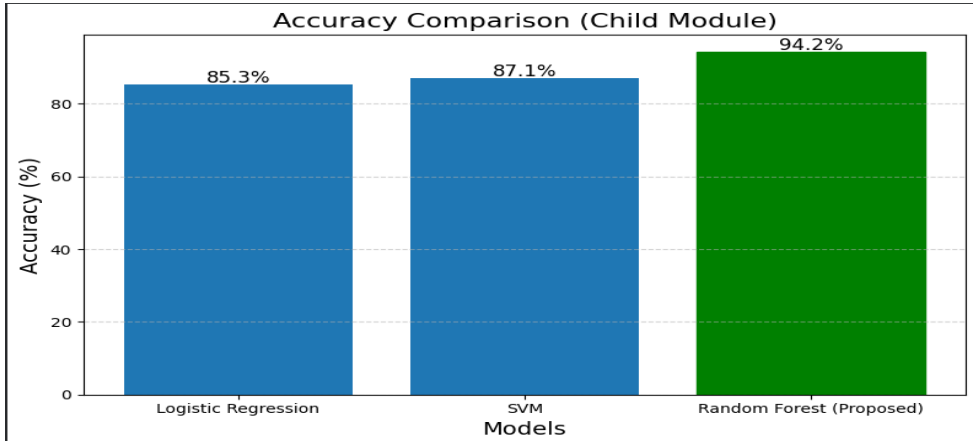


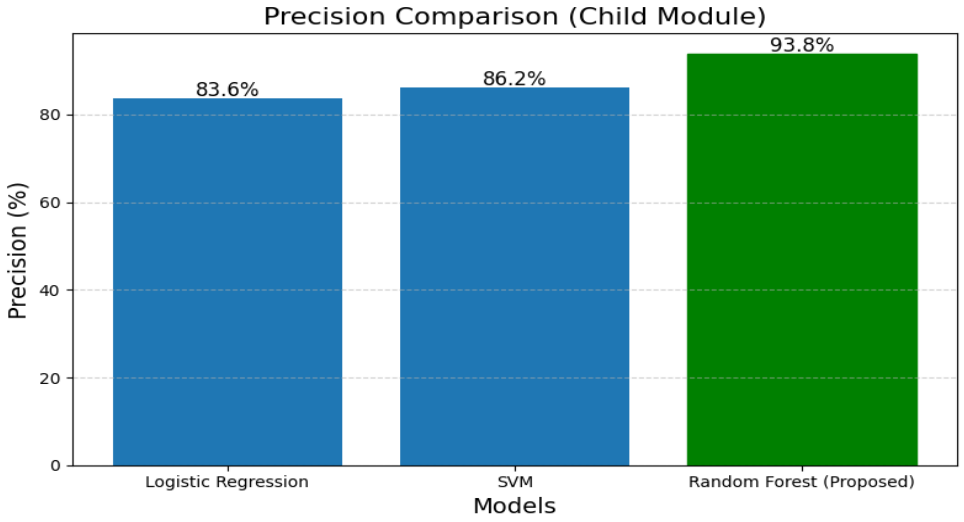


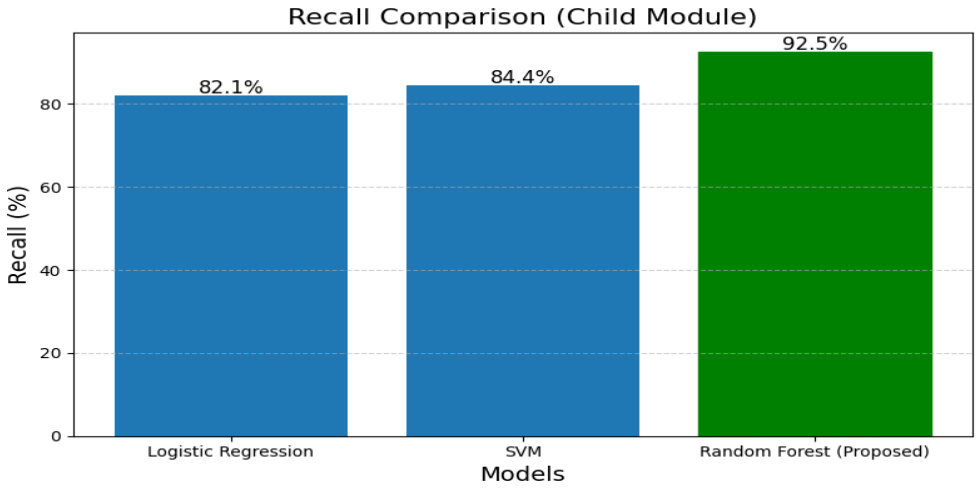
Results & Discussions

TABLE II *Questionnaire-Based ASD Prediction*

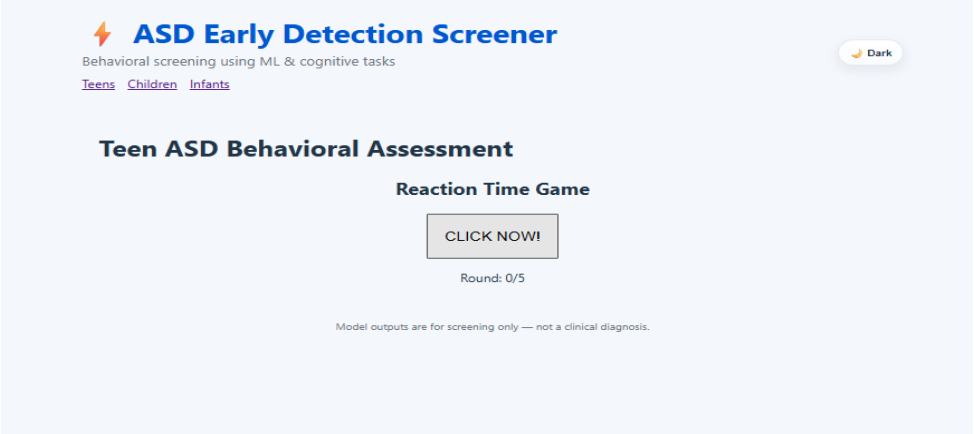
Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	85.3%	83.6%	82.1%	82.8%
SVM	87.1%	86.2%	84.4%	85.3%
Random Forest (Proposed)	94.2%	93.8%	92.5%	93.1%

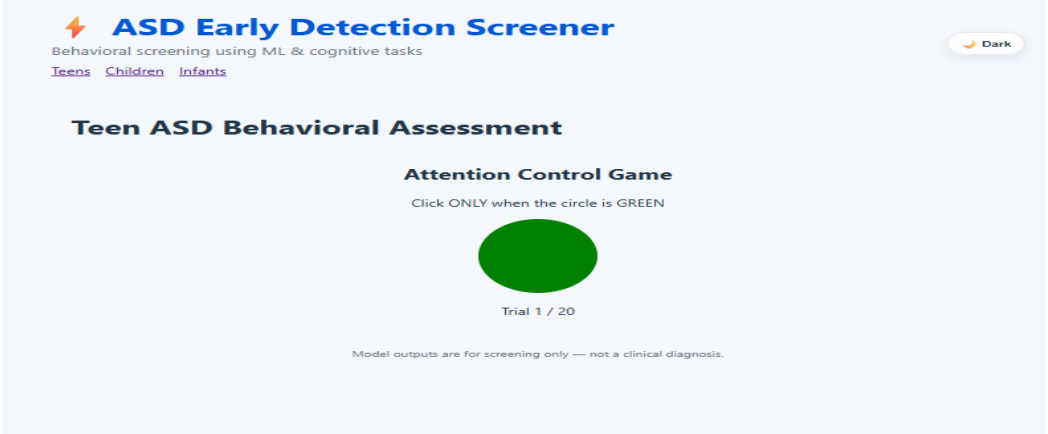




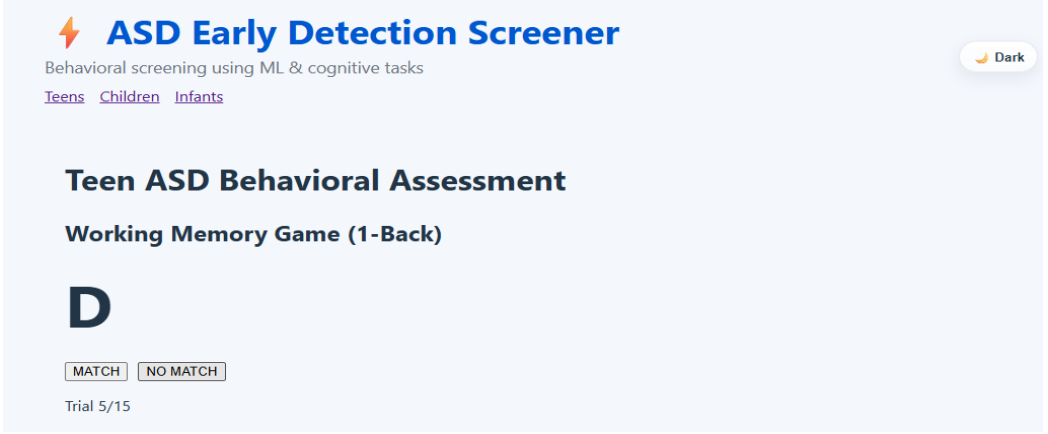


Results & Discussions

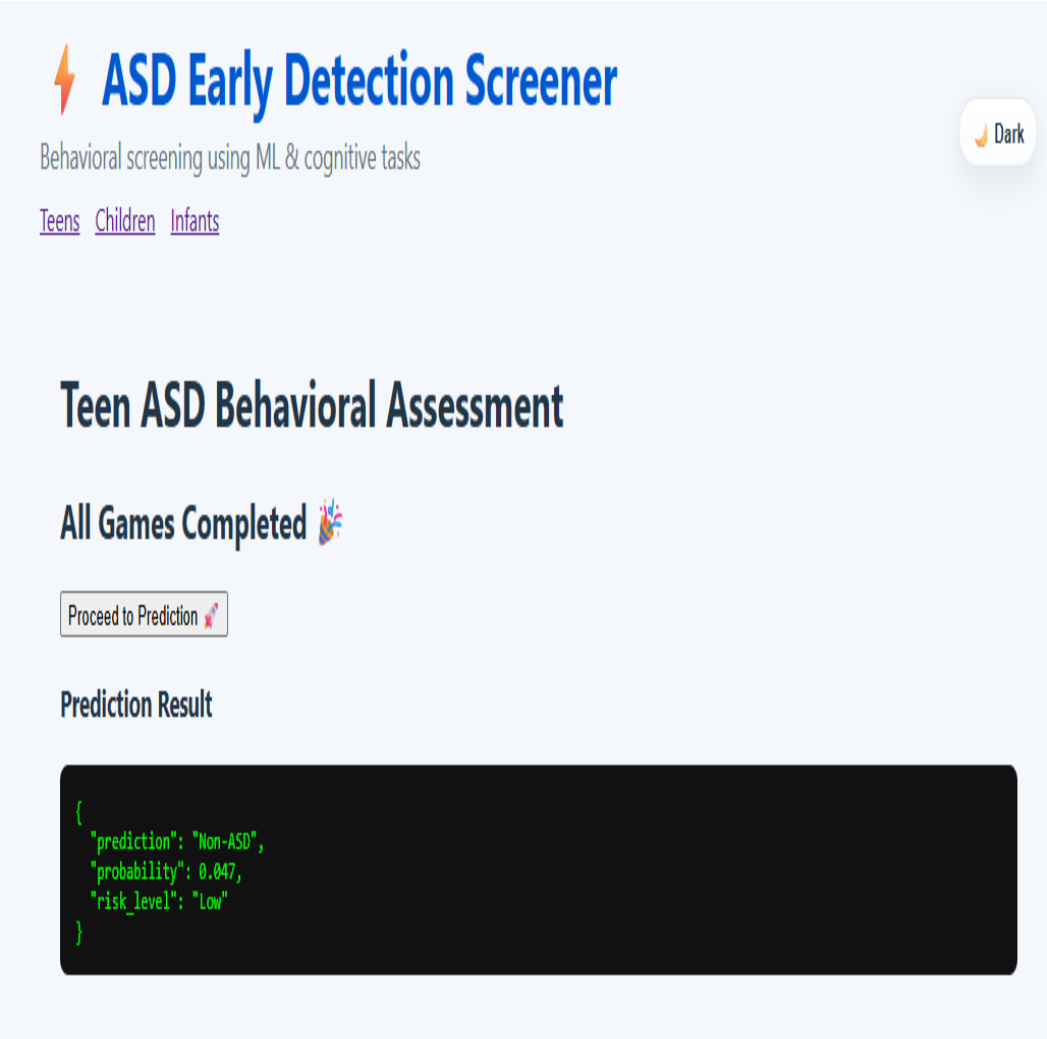


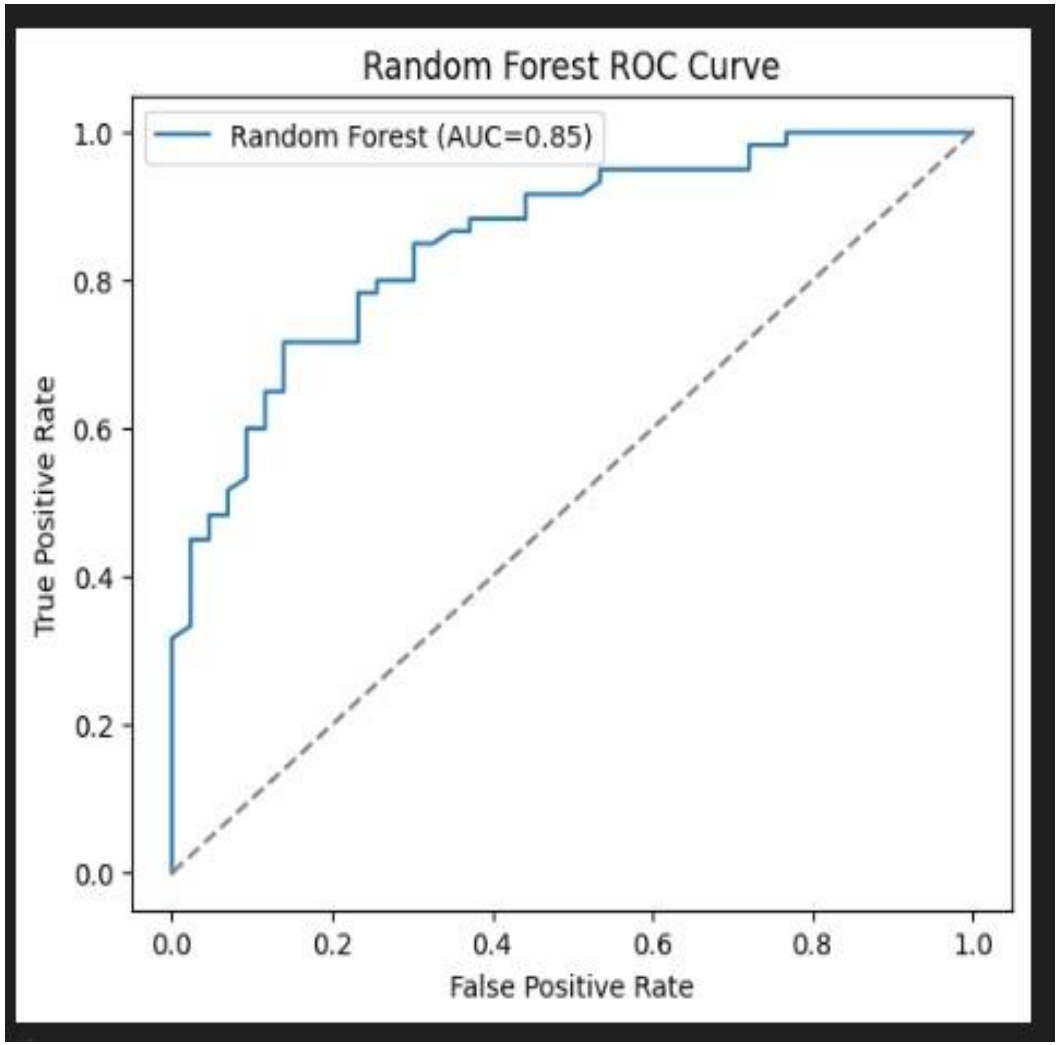






Results & Discussions

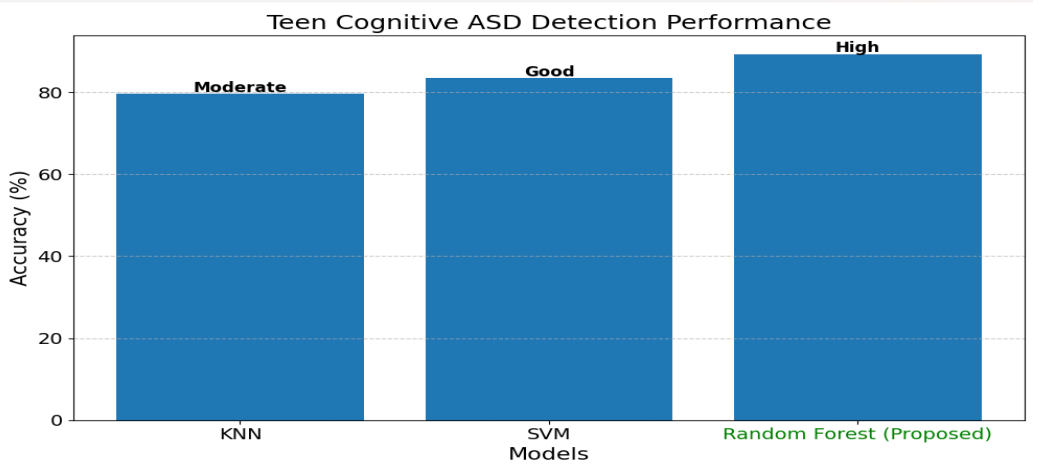


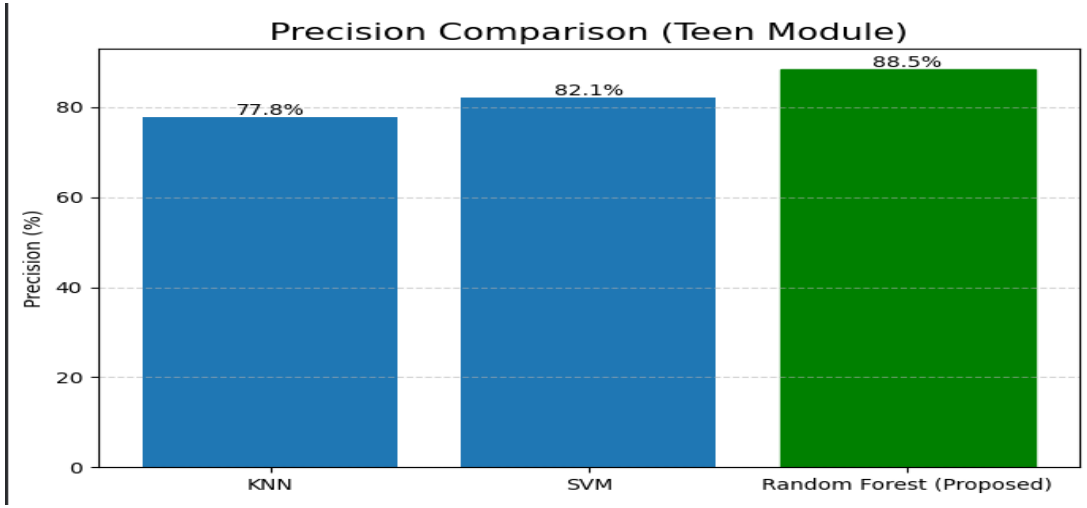


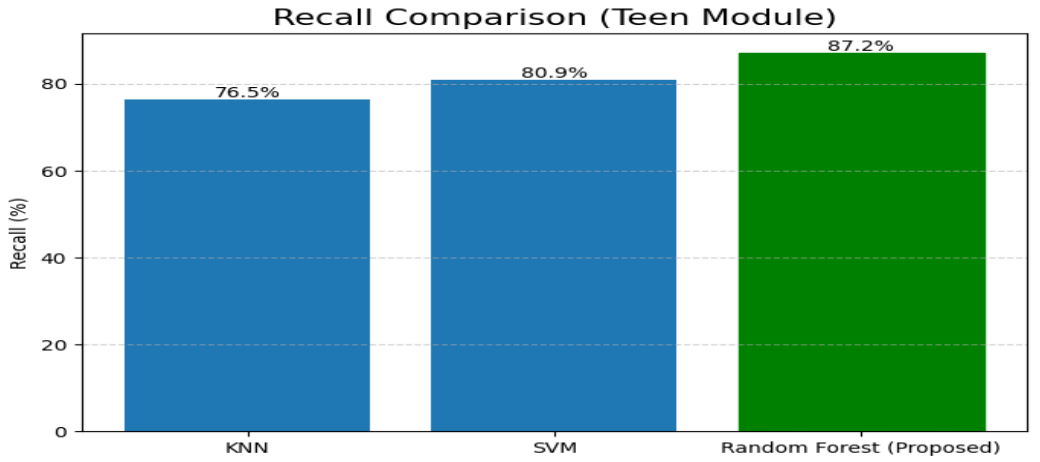
Results & Discussions

TABLE III *Teen Cognitive Feature Classification*

Model	Accuracy	Precision	Recall	Detection Quality
KNN	79.6%	77.8%	76.5%	Moderate
SVM	83.4%	82.1%	80.9%	Good
Random Forest (Proposed)	89.3%	88.5%	87.2%	High







Conclusion

- Developed a **multi-age ASD detection system** for **infants (0–3), children (4–12), teens (13–17)**.
- Combined **YOLO-based gaze tracking**, **LSTM time-series**, and **ML models** for reliable classification.
- Extracted key behavioral features:
  - Gaze deviation
  - Fixation duration
  - Saccade velocity
  - Pupil dilation
- Achieved strong detection performance across age groups.
- **SHAP Explainable AI** clarifies feature contributions → increases trust & interpretability.
- System demonstrates potential for **early screening** and **clinical decision support**.

Overall impact:

- Provides a **unified, multi-age ASD detection framework** not present in existing literature.

Future Work

1. Personalized Support Module

• Provide personalized therapy and intervention recommendations based on the predicted ASD severity level.

2. Mobile App Deployment

• Deploy lightweight YOLO And LSTM models in a mobile application for real-time ASD screening using smartphone cameras.

REFERENCES

- M. S. Farooq, M. Riaz, And A. Ali, "**Detection Of Autism Spectrum Disorder In Children Using Federated Machine Learning Techniques**," *Scientific Reports*, Vol. 13, 2023.
- R. A. Rasul Et Al., "**An Evaluation Of Machine Learning Approaches For Early Autism Detection,**" *Journal Of Biomedical Informatics*, 2024.
- R. Simeoli And L. Rossi, **"Using Machine Learning For Motion Analysis To Early Detect Autism Spectrum Disorder,"** *Applied Artificial Intelligence*, 2024.
- S. S. Rajagopalan Et Al., **"Machine Learning Prediction Of Autism Spectrum Disorder Using Clinical And Developmental Features,"** *JAMA Network Open*, Vol. 7, No. 4, 2024.
- N. Wankhede Et Al., **"Artificial Intelligence For Diagnosis And Treatment Of Autism Spectrum Disorder: A Review,"** *Artificial Intelligence In Medicine*, 2024.

Thank You!
Questions are welcome